CONSTRUCTIBILITY OF THE COHOMOLOGY SHEAVES OF RF_! FOR SEPARATED FINITELY PRESENTED MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Editorial note: the final reduction uses the first spectral sequence of the two-spectral-sequence lemma for a complex functor as a named input (the author also offers an alternative truncation argument); the per-term bookkeeping of the spectral sequence reduction is not written out.

Allowed standard prerequisites (editorial).

- Relative Mayer-Vietoris for lower shriek (Lemma lemma-relative-mayer-vietoris, canonical tag 0GKN). cited at proof line 4940. Together with the induction principle it reduces the claim from general X to affine X.
- Composition of lower shriek (Lemma lemma-shriek-composition, canonical tag 0F7J). cited at proof lines 4947-4948. Gives Rf_! = Rp_! after Ri_! for the factorization through the closed immersion.
- Constructibility for locally quasi-finite finitely presented morphisms (Lemma lemma-qf-f-shriek-constructible, canonical tag 0GK3). cited at proof lines 4949-4950. Handles the closed immersion part since Ri_! = i_! for a locally quasi-finite immersion.
- Boundedness of the derived lower shriek functors (Lemma lemma-derived-lower-shriek-bounded, canonical tag 0G2A). cited at proof lines 4961-4964. Rf_! has finite cohomological dimension, allowing K to be assumed bounded below.
- Two spectral sequences for a complex functor (Derived Categories, Lemma derived-lemma-two-ss-complex-functor, canonical tag 015J). cited at proof lines 4964-4966. Its first spectral sequence reduces the claim from bounded below K to cohomology sheaves of K.
- Constructibility for relative dimension 1 (Lemma lemma-constructible-shriek-rel-dim-1, canonical tag 0GKZ). cited at proof lines 4967-4968. The A^1 case target at which the reduction arrives.

Source reference. Original lemma 0GKN: sources/stacks-chow/tags.txt, source lines 20693–20693. Cited at proof line 4940: together with the induction principle it reduces the claim to affine X.

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Source reference. Original lemma 0F7J: sources/stacks-chow/tags.txt, source lines 19009–19009. Cited at proof lines 4947-4948: gives Rf_! = Rp_! after Ri_! for the factorization through the closed immersion.

Source reference. Original lemma 0GK3: sources/stacks-chow/tags.txt, source lines 20673–20673. Cited at proof lines 4949-4950: handles the closed immersion part since Ri_! = i_! for a locally quasi-finite immersion.

Source reference. Original lemma 0G2A: sources/stacks-chow/tags.txt, source lines 20050–20050. Cited at proof lines 4961-4964: Rf_! has finite cohomological dimension, so K may be assumed bounded below.

Source reference. Original lemma 015J: sources/stacks-chow/tags.txt, source lines 1510–1510. Cited at proof lines 4964-4966: its first spectral sequence reduces the claim from bounded below K to the cohomology sheaves of K.

Source reference. Original lemma 0GKZ: sources/stacks-chow/tags.txt, source lines 20704–20704. Cited at proof lines 4967-4968: the A^1 case target at which the reduction arrives.

Theorem 0.2. *Let $f : X \to Y$ be a separated morphism of finite presentation of quasi-compact and quasi-separated schemes. Let Λ be a Noetherian ring. Let K be an object of $D^+_{tors,c}(X_{étale}, \Lambda)$ or of $D_c(X_{étale}, \Lambda)$ in case Λ is torsion. Then $Rf_!K$ has constructible cohomology sheaves, i.e., $Rf_!K$ is in $D^+_{tors,c}(Y_{étale}, \Lambda)$ or in $D_c(Y_{étale}, \Lambda)$ in case Λ is torsion.*

Proof. The question is local on Y hence we may and do assume Y is affine. By the induction principle and Lemma 0GKN we reduce to the case where X is also affine.

Assume X and Y are affine. Since X is of finite presentation, we can choose a closed immersion $i : X \to \mathbf{A}^n_Y$ which is of finite presentation. If $p : \mathbf{A}^n_Y \to Y$ denotes the structure morphism, then we see that $Rf_! = Rp_! \circ Ri_!$ by Lemma 0F7J. By Lemma 0GK3 we have the result for $Ri_! = i_!$. Hence we may assume f is the projection morphism $\mathbf{A}^n_Y \to Y$. Since we can view f as the composition

$$X = \mathbf{A}^n_Y \to \mathbf{A}^{n-1}_Y \to \mathbf{A}^{n-2}_S \to \ldots \to \mathbf{A}^1_Y \to Y$$

we may assume $n = 1$.

Assume Y is affine and $X = \mathbf{A}^1_Y$. Since $Rf_!$ has finite cohomological dimension (Lemma 0G2A) we may assume K is bounded below. Using the first spectral sequence of Derived Categories, Lemma 015J (or alternatively using an argument with truncations), we reduce to showing the result of Lemma 0GKZ. □